

Black and White Holes Do Not Host Universes: A Structural Refutation of Interior-Cosmology Under Triadic Closure

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Abstract

Popular science frequently suggests that universes may reside inside black holes, that the Big Bang may correspond to a white-hole interior, or that cosmological inflation may arise from interior geometric expansion. These interpretations rely on classical general-relativistic interior solutions or speculative extensions of holography. This paper provides a structural refutation of interior-cosmology using the architectural framework developed across the Triadic Cosmos ecosystem. Building on *Closed Ecosystems and the Limits of Local Closure*, *Unifying GR and QM through Triadic Closure*, and *The Triadic Cycle with AdS/CFT Recursion*, we show that black-hole interiors lack independent informational, energetic, and geometric degrees of freedom; do not support triadic recursion; do not contain emergent temporal structure; and therefore cannot host universes. White holes, meanwhile, are overcomplete GR geometries that fail triadic closure and cannot instantiate causal or dynamical domains. Universes require active triadic closure, horizon-bounded recursion, and non-zero informational capacity. Neither black-hole nor white-hole interiors satisfy these conditions. We conclude that interior-cosmology is architecturally impossible and that recursive black-hole multiverse models cannot be physically realized.

1 Introduction

The idea that universes may exist inside black holes, or that the Big Bang corresponds to a white-hole interior, has gained traction in popular science. Classical GR admits interior solutions that resemble expanding regions, and speculative models propose that white holes may eject entire universes. These interpretations arise from geometric extrapolation rather than structural analysis.

The Triadic Cosmos ecosystem provides a different perspective. Earlier work established that physical reality is organized by horizon-bounded closures whose triadic recursion ($E \rightarrow G \rightarrow I \rightarrow E$) generates emergent time, causal structure, and informational accessibility. Black-hole interiors were shown to lack independent triadic roles, causing recursion to terminate at the horizon. White holes were shown to be overcomplete geometries that cannot instantiate physical domains.

This paper unifies these insights into a single structural refutation of interior-cosmology.

2 Background: Triadic Closure and Interior Structure

2.1 Triadic Closure

Triadic closure is the minimal architectural mechanism linking energy (E), geometry (G), and information (I). A physical domain exhibits triadic closure if:

- informational capacity is non-zero,

- energetic and geometric roles satisfy causal accessibility,
- holographic embedding is consistent,
- the recursive update operator R is well-defined,
- the full triadic cycle closes without contradiction.

Emergent time arises from successive triadic updates. A domain without triadic recursion has no temporal structure and cannot host physical evolution.

2.2 Black-Hole Interiors

The Triadic Cycle with AdS/CFT Recursion established that black-hole interiors lack independent informational, energetic, and geometric degrees of freedom. The triadic recursion terminates at the horizon:

“The interior does not support emergent temporal structure and is best understood as a holographic shadow of boundary data rather than a dynamical region.”

Thus, black-hole interiors are non-recursive domains with no time, no causal structure, and no informational capacity.

2.3 White Holes as Overcomplete Geometries

Unifying GR and QM through Triadic Closure showed that GR admits geometries that are mathematically valid but architecturally impossible. White holes are among these:

- they lack informational embedding,
- they violate causal accessibility,
- they cannot support energetic evolution,
- they fail triadic closure.

White holes are therefore non-physical structures.

3 Why Universes Require Active Triadic Closure

A universe is a closure: a horizon-bounded domain with active triadic recursion. It requires:

- informational capacity (I),
- energetic dynamics (E),
- geometric causal structure (G),
- recursive update operator R generating emergent time.

Without these roles, no physical evolution can occur. A universe cannot exist in a domain without triadic closure.

4 Holography and Boundary Misinterpretations

4.1 Black Holes Boundaries

A common objection to the claim that black holes cannot host universes is based on holography: if a 3D bulk can emerge from a 2D boundary, then perhaps a universe could exist on or within the horizon of a black hole. This interpretation misunderstands the role of the horizon in holographic architectures.

In holographic dualities such as AdS/CFT, the 2D boundary encodes the full dynamical degrees of freedom of the 3D bulk. Crucially, the boundary is *outside* the bulk region it encodes. The horizon of a black hole does not play this role. Instead:

- the horizon encodes the *exterior* region, not the interior,
- the interior is a *holographic shadow* of boundary data,
- the interior has no independent informational degrees of freedom,
- the interior has no independent energetic or geometric evolution,
- the interior does not support triadic recursion or emergent time.

As established in *The Triadic Cycle with AdS/CFT Recursion*, the interior of a black hole is not a dynamical domain but a recursion-halted region. The holographic encoding on the horizon reconstructs the exterior geometry; it does not generate a new interior universe. The 2D boundary of a black hole cannot host a universe because it does not encode an autonomous 3D bulk. It encodes only the exterior causal domain.

Therefore, holography does not provide a mechanism for universes to exist inside black holes. The horizon is not a universe-bearing boundary, and the interior is not a dynamical bulk.

4.2 White Holes Boundaries

White holes are sometimes invoked as potential “universe generators,” with the suggestion that a universe might emerge from a white-hole boundary in a holographic manner. This interpretation fails for structural reasons.

White holes are overcomplete GR geometries that lack triadic closure. They possess:

- no causal accessibility,
- no informational embedding,
- no energetic evolution,
- no recursive triadic updates.

Even if one attempts to apply holography, the boundary of a white hole cannot encode a physical bulk. Holography requires:

- a well-defined boundary theory,
- non-zero informational capacity,
- causal accessibility of degrees of freedom,
- consistent energetic evolution,
- a recursive update structure supporting emergent time.

White holes satisfy none of these conditions. Their “boundary” is not a holographic surface in the AdS/CFT sense; it does not encode a dynamical bulk and cannot generate a universe. Instead, white holes represent non-physical GR artifacts filtered out by triadic closure.

Thus, holography cannot rescue white holes as universe-hosting structures. Their boundaries do not encode autonomous 3D domains, and their interiors do not support any form of physical evolution.

5 Why Black Holes Cannot Host Universes

5.1 No Informational Capacity

Black-hole interiors have no independent informational degrees of freedom. All information is encoded holographically on the horizon. A universe requires internal informational structure.

5.2 No Energetic Dynamics

Energetic roles inside the horizon are not independent. The interior does not support Hamiltonian evolution or energetic reconfiguration. A universe requires energetic dynamics.

5.3 No Geometric Causal Structure

Interior geometry is not an accessible causal domain. It is a holographic shadow of boundary data. A universe requires geometric causal structure.

5.4 No Triadic Recursion

Triadic recursion halts at the horizon. Without recursion, there is no emergent time. A universe requires temporal evolution.

5.5 No Closure

A black-hole interior is not a closure. It is a non-recursive region. A universe is a closure. Therefore, a universe cannot exist inside a black hole.

6 Why White Holes Cannot Host Universes

6.1 White Holes Are Overcomplete GR Geometries

White holes are mathematically permissible but architecturally impossible. They lack:

- informational capacity,
- energetic accessibility,
- causal structure,
- triadic recursion.

6.2 No Causal Domain

White holes violate causal accessibility: no observer can enter, no information can be embedded, and no dynamical evolution can occur.

6.3 No Energetic or Informational Evolution

White holes cannot support Hamiltonian evolution or informational change. They cannot host physical processes.

6.4 No Triadic Closure

White holes fail all closure conditions. They cannot instantiate emergent time or recursive triadic updates. Therefore, they cannot host universes.

7 Unified Refutation of Interior-Cosmology

Interior-cosmology requires:

- interior informational capacity,
- interior energetic dynamics,
- interior geometric causal structure,
- interior triadic recursion.

Black-hole interiors satisfy none of these conditions. White holes satisfy none of these conditions. Therefore, interior-cosmology is architecturally impossible.

8 Conclusion

Black and white holes do not host universes. Their interiors lack informational, energetic, and geometric roles; do not support triadic recursion; do not contain emergent time; and are not closures. Universes require active triadic closure, which neither black-hole nor white-hole interiors can instantiate. Recursive black-hole multiverse models are architecturally impossible.

A Triadic Cosmos Ecosystem Context

This paper is part of the *Triadic Cosmos* ecosystem, which develops a unified conceptual and computational framework spanning physical ontology and modular architectures for intelligent systems. All materials, extended notes, and evolving documentation are available in the public repository.

<https://github.com/triadic-cosmos>.

References

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